INCH-POUND

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SUPERSEDING
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2 April 1974

COMMERCIAL ITEM DESCRIPTION

CUTTER, BARBED WIRE, INSULATED HANDLES, HAND OPERATED

The General Services Administration has authorized the use of this Commercial Item Description for all federal agencies.

1. SCOPE

1.1 Scope. This commercial item description covers one type of hand operated, barbed wire cutter with insulated handles.

1.2 Intended use. The cutters covered by this commercial item description are intended for use in the field to cut barbed wire which is charged with a high voltage.

2. SALIENT CHARACTERISTICS

2.1 Materials and components. Materials and components shall be as specified herein. Where materials are not definitely specified, they shall be of the quality normally used for this purpose in commercial practice provided the completed cutters comply with all provisions of this commercial item description.

2.2 Tool steel, alloy. Alloy tool steel shall conform either to ASTM A 681 or commercial fine grain tool steel possessing the chemical composition specified in table I.

Beneficial comments, recommendations, additions, deletions, clarifications, etc., and any other pertinent data which may improve this document should be sent to: Commander, U.S. Army Natick Research, Development and Engineering Center, ATTN: SSCNC-WEF, Natick, MA 01760-5018 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 5110

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TABLE I. Tool steel chemical composition

Carbon	Silicon	Vanadium	Manganese	Phosphorus and sulfur
0.61 - 0.69	0.75 - 1.10	0.15 - 0.25	0.30 - 0.50	0.025 max.

2.3 Design and construction. The barbed wire cutter shall be the manufacturer's standard commercial product similar, but not necessarily restricted to the design of the cutter shown in figure 1. The cutter shall consist of a pair of jaws, insulated handles, a strap, rivets and pivot bolts. The overall length of the cutter shall be 6 inches ± 0.25 inches.

2.3.1 Jaws. The jaws shall be made from alloy tool steel specified in 2.2. The bevel of the finished jaws shall have a Rockwell hardness on the Rockwell C scale of not less than 53 HRC nor greater than 61 HRC, and shall be verified by the test specified in 4.2.1. The tool shall have means for adjusting cutting edges. The space between cutting edges shall be not more than 0.004 inches.

2.3.2 Strap and handles. The strap and handles shall be fabricated from a commercial grade carbon steel in accordance with the manufacturer's standard practice. The strap and finished handles shall have a Rockwell hardness on the Rockwell C scale of not less than 43 HRC nor greater than 47 HRC, and shall be verified by the test specified in 4.2.1.

2.3.3 Rivets and pivot bolts. Rivets and pivot bolts shall be fabricated from a commercial grade carbon steel in accordance with the manufacturer's standard practice. Before assembly, rivets and pivot bolts shall have a Rockwell hardness on the Rockwell C scale of not less than 30 HRC nor greater than 40 HRC, and shall be verified by the test specified in 4.2.1.

2.3.4 Handle insulation. The handle insulation shall be black and capable of being molded to a steel handle. The insulation shall have:

(1) A dielectric strength to withstand a 14,000 volt RMS, 60 hertz, AC potential for 1 minute, with no leakage or passage of current in excess of 70 microamperes per inch of immersion per cutter handle in a saline solution when tested as specified in 4.2.2. The insulation shall show no abnormal distortion or blistering after testing in accordance with 4.2.2.

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(2) Resistance to moisture absorption so that after immersion in water, the insulation shall withstand a 7,000 volt RMS, 60 hertz, AC potential, with no flash over (surface discharge), spark over (air discharge), or breakdown (puncture discharge), when tested as specified in 4.2.3.

(3) Resistance to impact so that the insulation of the handles shall withstand the blow of a sharp edged object and also comply with the distortion, blistering, and dielectric requirements in (1) above, when tested as specified in 4.2.4.

(4) Resistance to heat, acid, and alkali with no deterioration of the handle insulation such as blistering, swelling, peeling, becoming spongy, or being dissolved, when tested as specified in 4.2.5 and 4.2.6 as applicable.

2.4 Performance. When tested as specified in 4.2.7, the cutter shall make clean cuts with spring wire at all points between the innermost and outermost points of the cutting edges without: (1) Binding of movable parts; (2) Cutter parts becoming loose; (3) Indenting or chipping of the jaw edges; (4) Permanent deflection or deformation of the jaw and handles.

2.5 Finish application. Exposed metal surfaces of the cutter shall have a black, corrosion resistant finish in accordance with the manufacturer's standard practice. Prior to the application of finish, the cutter surfaces shall be treated or cleaned to provide a smooth and adherent surface.

2.6 Metric products. Products manufactured to metric dimensions will be considered on an equal basis with those manufactured using inch-pound units, provided they fall within the tolerances specified using conversion tables contained in the latest revision of ASTM E 380, and all other requirements of this Commercial Item Description are met. If a product is manufactured to metric dimensions and those dimensions exceed the tolerances specified in the inch-pound units, a request should be made to the contracting officer to determine if the product is acceptable. The contracting officer has the option of accepting or rejecting the product.

2.7 Workmanship. The completely fabricated cutter shall be free from burrs, slivers, and sharp edges. Cutting edges shall be parallel, and in alignment. Parts shall be as specified, free from dents, fractures, breakage or deformation. No parts shall be damaged or impaired in any way and metal parts shall show no evidence of rust or corrosion. There shall be no defects which impair the durability, appearance, or serviceability of the item.

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3. REGULATORY REQUIREMENTS

3.1 Contractor recovered materials. The offeror/contractor is encouraged to use recovered materials to the maximum extent practicable, in accordance with paragraph 23.403 of the Federal Acquisition Regulation (FAR).

4. QUALITY ASSURANCE PROVISIONS

4.1 Product conformance. The products provided shall meet the salient characteristics of this commercial item description, shall conform to the producer's own drawings, specifications, standards, and quality assurance practices, and be the same product that has been sold in the commercial marketplace for over one year, or is the same product that has successfully been delivered to the Government on a previous contract or purchase order. The Government reserves the right to require proof of such conformance prior to first delivery and thereafter as may be otherwise provided for under the provisions of the contract. The following market acceptability criteria are necessary to document the quality of the product to be provided under this CID:

a. The company producing the item must have been producing a product meeting the requirements of this CID for at least 2 years.

b. The company must have sold 1000 items meeting this CID, in the commercial or government marketplace over the past 2 years.

4.2 Tests.

4.2.1 Hardness. Test the hardness of the jaws, rivets, pivot bolts, straps and handle in accordance with ASTM E 18 to determine compliance with the hardness requirements of 2.3.1 through 2.3.3, as applicable. Base the hardness on the average of three readings. Test the hardness of the jaws on the bevel, 0.125 inches from the cutting edge. Test the hardness of the rivets, pivot bolts, straps and handle at any location on the surface.

4.2.2 Dielectric strength. Immerse the insulated handle of the cutter in 1 percent salt water solution, 2 to 2.5 inches from the top of the insulation. The metal part of the handle shall be one electrode and the salt solution shall be the other electrode, preferably as the ground terminal. The potential between the electrodes shall be raised at a rate of 3,000 volts per second until the final test voltage of 14,000 RMS volts, 60 hertz, AC is reached. Maintain the test voltage for a period of 1 minute. During this period, voltage and current readings shall be observed to determine conformance to requirements of 2.3.4 (1).

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The test equipment shall have a capacity of not less than 1 kilovolt ampere.

4.2.3 Resistance to moisture absorption. Immerse the insulated handle in water for a period of 30 minutes, then wipe dry with a cloth and immediately test as follows. Place two metal foil electrodes on the insulated portion of the handle so their inner edges are 1 inch apart. A potential of 7,000 RMS volts peak, 60 hertz, AC shall be applied to the electrodes at a rate of 3,000 volts per second. Apply the test voltage for a period of 30 seconds to determine compliance with 2.3.4 (2).

4.2.4 Resistance to impact. Rigidly mount the insulated handle. Swing a hardened steel cutting edge, suspended as a pendulum, to impart a blow to the handle insulation, equivalent to 5 foot pounds of energy. The cutting edge shall strike the insulation at right angles to the longitudinal axis of the handle to determine compliance with 2.3.4 (3). The cutting edge shall consist of a 90 degree wedge, 0.010 inches in width and 1 inch long.

4.2.5 Resistance to acid. Immerse and soak the insulated handle in a 20 percent sulfuric acid solution for a period of 1 hour at 150°F minimum; then remove from the acid solution and wipe dry. The insulation shall be examined to determine compliance with the applicable requirements of 2.3.4 (4).

4.2.6 Resistance to alkali. Immerse and soak the insulated handle in a 40 percent sodium hydroxide solution for a period of 1 hour at 150°F minimum; then remove from the alkali solution and wipe dry. The insulation shall be examined to determine compliance with the applicable requirements of 2.3.4 (4).

4.2.7 Performance. Make 10 complete cuts with the cutter on a commercially available carbon hard-drawn spring wire, 0.116 inches in diameter. Make each cut at separate points from the innermost to the outermost points of the cutting edges to determine compliance with the performance requirements of 2.4.

5. PACKAGING

5.1 Packaging. Preservation, packing, and marking shall be as specified in the contract or purchase order (see 6.1).

6. NOTES

6.1 Ordering data. Acquisition documents must specify the following:

a. Title, number and date of this document.
b. Packaging requirements (see 5.1).

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6.2 National Stock Number. The following national stock number corresponds to this CID: 5110-00-595-8229.

6.3 Sources for non-Government association documents.

ASTM

A 681	-	Standard Specification for Tool Steels Alloy
E 18	-	Methods of Tests for Rockwell Hardness and Rockwell Superficial Hardness of Metallic Materials
E 380	-	Use of the International System of Units (SI) (the Modernized Metric System)

ASTM documents are available from:

The American Society for Testing and Materials (ASTM)
100 Barr Harbor Drive
West Conshohocken, PA 19428

6.4 Sources of supply. The manufacturer whose products are known to meet the CID requirements is listed below. However, competition is not limited to this company:

Ameritool
Bar-Es Division
16741 West Park Circle Drive
Chagrin Falls, OH 44023

MILITARY INTERESTS:

Custodians

Army - GL
Navy - AS
Air Force - 99

Review activities

Navy - CG, MC
Air Force - 03, 84

CIVIL AGENCY COORDINATING ACTIVITY:

DOI - BLM
GSA - FSS

PREPARING ACTIVITY:

Army - GL

Project No. 5110-0349

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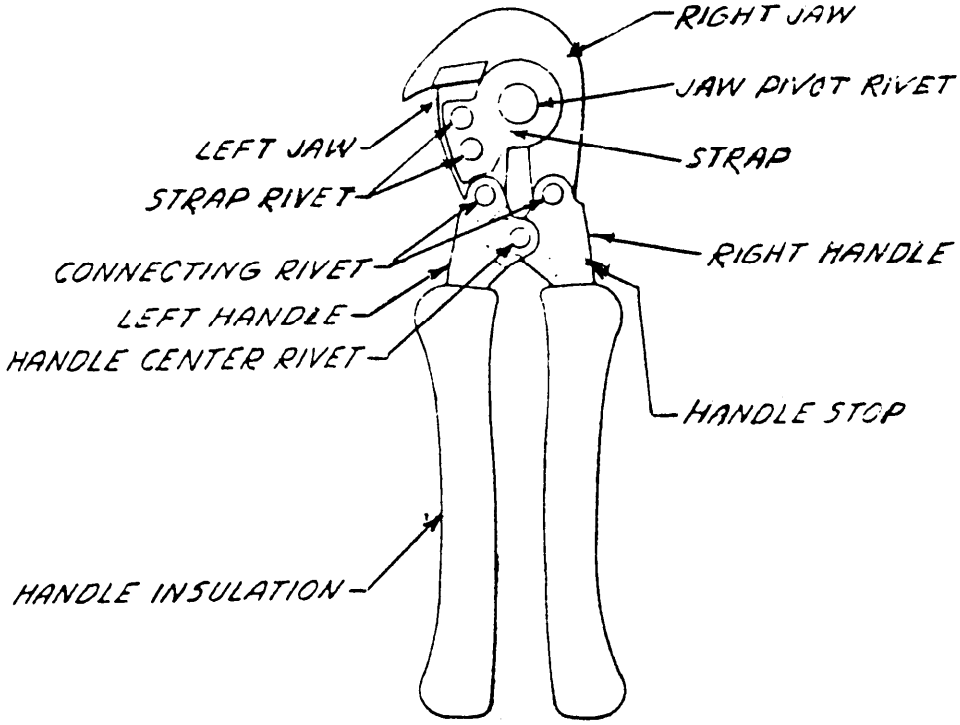


Figure 1. Cutter, Barbed Wire, Insulated Handles, Hand Opperated.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER A-A-52207	2. DOCUMENT DATE (YYMMDD) 960812

3. DOCUMENT TITLE
Cutter, Barbed Wire, Insulated Handles, Hand Operated

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE (Include Area Code) (1) Commercial	(2) AUTOVON
U.S. Army Natick RD&E Center	508-233-5175	256-5175
c. ADDRESS (Include Zip Code) Commander, U.S. Army Natick RD&E Center ATTN: SSCNC-WEF Natick, MA 01760-5018	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

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